University of Bahrain
College of Science
Postgraduate Programs

Course Project

Information

The Master Program	M.Sc. in Big Data Science and Analytics
Course Title & Code	Statistical Data Analysis BDSA 602
Academic Semester	First Semester 2025 – 2026
Instructor's Name	Dr. Sawsan Hilal

DEADLINE AND MARK WEIGHT

Please note that the due date for your project report is **THURSDAY 18TH DECEMBER 2025**.

The companion presentation will take place on **TUESDAY 16TH DECEMBER 2026**.

The maximum mark for this project report is **20 points** as detailed on the last page.

The maximum mark for the presentation is **10 points** as detailed on the last page.

PROJECT STATEMENT

The purpose of this project is to conduct a research study based on the theoretical knowledge and practical skills with R programming language acquired from the Statistical Data Analysis course with a specific interest in the first four chapters. This project is mainly on supervised learning, specifically, it focuses on classification problems. Students will work on this project in groups. They are requested to choose a specific problem that can be handled using classification techniques and to collect the relevant dataset to be analyzed to solve the problem of concern.

PROJECT REPORT STRUCTURE

The project report should be prepared as a **conference paper** with a maximum of **six pages**.

1. FORMAT

Students need to use Project Report Template (attached with these guidelines).

2. TITLE

Students need to choose a title that reflects their study.

3. ABSTRACT

The abstract provides a **short summary** of the research study, and it should be written as a paragraph that covers an introduction to the topic of concern, a clear statement about the problem to be addressed, the implemented methods, and the study main findings. This section is to be written once you complete the study.

4. INTRODUCTION

This section should start with a motivational paragraph that highlights the **background** of the problem to be addressed and provides a brief **literature review** in relation to the problem of concern. The section ends with an outline of the adopted approach for the current study.

5. EXPLORATORY DATA ANALYSIS

This section introduces the reader to the relevant dataset stating its source and characteristics. It is recommended to divide this section into subsections that cover different aspects of EDA such as Data Collection, Data Processing, and Data Summary.

6. METHODOLOGY

This section describes the implemented methods which should be written in good detail. Students are requested to adopt the listed methods, to briefly explain each method, and to provide references for more details on the selected methods for interested readers.

- Logistic Regression
- Linear Discriminant Analysis.
- Quadratic Discriminant Analysis.
- KNN classifier with the optimal parameter K.

7. RESULTS AND DISCUSSION

This section details the results of the research study which should be reported in terms of tables and/or figures with proper captions and text commentary. Moreover, the section should provide a good discussion for the results with logical explanation and should highlight the similarities and contrasts with other relevant studies that have been used in the introduction/literature review.

8. CONCLUSION

This section summarizes the conducted research study with its main findings and recommendations as well as directions of improvement.

9. REFERENCES

The references used in the literature review should be properly listed in order according to their appearance in the paper. The references should follow **APA reference style**.

SUBMISSION FOLDER

The folder to be submitted should contain the following items:

- **Study Report** (as pdf)
- **Presentation Slides**
- **R script** (in word)
- **Study Dataset**

DATASET SOURCES

Use existing datasets from dataset repositories after approval from the instructor. Here are examples of these dataset repositories:



UCI ➔ Machine Learning Repository
https://archive.ics.uci.edu/ml/index.php



KDnuggets ➔ Datasets for Data Mining, Data Science, and Machine Learning
https://www.kdnuggets.com/datasets/index.html



kaggle ➔ Machine Learning and Data Science Community
https://www.kaggle.com/



KDD Cup Achieves ➔ The community for data mining, data science and analytics
https://kdd.org/kdd-cup

ASSESSMENT CRITERIA

Project Report

Criterion	Mark Weight
ABSTRACT A brief introduction to the problem to motivate the work.	01
INTRODUCTION A brief introduction to the problem to motivate the work.	04
EXPLORATORY DATA ANALYSIS A good description of data to be analysed with numerical and graphical summary.	04
METHODOLOGY A good description for the implemented statistical learning methods (implemented models).	04
RESULTS AND DICUSSION An organized report for the results of the conducted statistical analysis along with their interpretations in the context of the problem.	04
CONCLUSION An informative conclusion for the analysis highlighting the best model with a justification.	01
REFERENCES A list of the references used to complete the required tasks with a proper referencing style.	01
OTHERS The writing skills: clarity of writing, correct grammar, etc.	01
Total	20

Project Presentation

This is a 15-minute presentation about your project to be given in the class on the specified date.

Criterion	Mark Weight
INTRODUCTION A brief introduction to the problem to motivate the work.	01
EXPLORATORY DATA ANALYSIS A good description of data to be analysed with numerical and graphical summary.	01
METHODOLOGY A concise presentation for the adopted statistical methods/models to solve the problem of concern with a justification for their implementation.	03
RESULTS AND DISCUSSION An organized report for the study main findings along with their interpretations so to provide discussion in the context of the problem.	03
CONCLUSION An informative conclusion for the analysis highlighting the best model with a justification.	01
OTHERS The presenting skills and answering questions.	01
Total	10